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MILITARY SPECIFICATION

SHELL, ILLUMINATING, M314A1 AND M314A1B1 FOR 105-MM HOWITZERS, M2 AND M4 LOADING, ASSEMBLING AND PACKING

This specification has been approved by the Department of Defense for the use of the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers loading, assembling and packing of M314A1, and M314A1B1 Illuminating Shell for use in 105-mm Howitzers, M2 and M4.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, drawings, and publications, and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-P-223 — Powder, Black.

MIL-G-2550 — General Specification for Ammunition Except Small Arms Ammunition.

MIL-P-20464 — Pyrotechnics: Method of Determination of Candlepower and Color Value.

DRAWINGS

ORDNANCE CORPS

75-14-508 — Shell, Illuminating, For 105MM, M314A1B1, Loading Assembly.

PUBLICATIONS

ARMY ORDNANCE CORPS

ORD-SIP-M18-314— Shell, Illuminating, 105MM, M314A1 and M314A1B1.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—Materials and parts shall be in accordance with applicable specifications and drawings.

3.2 The loading assembly shall comply with all requirements specified on Drawing 75-14-508, and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 Parachute assembly. — The parachute assembly shall comply with all requirements specified on the drawings and in applicable specifications and shall be sewn and assembled in accordance with good commercial practice for textile products. Any canopy with more than one hole exceeding 1/8 inch in diameter or any canopy with more than one patch 1 inch in diameter shall be classed as defective. Dirt, oil or rust spots not ex-

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ceeding ¼ inch in diameter will be permitted on canopy. However, a total of more than two of such spots will cause the assembly to be classed defective.

3.4 **Moisture content.**

3.4.1 *Illuminant composition.* — The moisture content of the sodium nitrate of the illuminant composition at the blending operation at the time of blending shall not exceed 0.10 percent when determined as specified in 4.4.1.

3.4.2 *First fire composition.*—The moisture content of the barium nitrate and zirconium hydride of the first fire composition at the blending operation at the time of blending shall not exceed 0.10 percent when determined as specified in 4.5.1.

3.4.3 *Black powder.*—The moisture content of the black powder at the loading station at the time of loading the charge assembly shall be as specified in Specification JAN–P–223 when determined as specified in 4.6.

3.5 **Proving ground requirements for the shell.**

3.5.1 *Safety.*—There shall be no premature bursts or evidence thereof when tested as specified in 4.8. (A "premature burst" shall be defined as a functioning of the shell in the bore of the weapon or in the first 100 yards of flight).

3.5.2 *Functioning.* — A score of not less than 75 percent shall be obtained, based on a normal shell which functions properly, providing effective illumination for not less than 43 seconds, when tested as specified in 4.8.

3.6 **Repainting.**—The painting and markings shall be restored by the assembling contractor on any components furnished to him painted and marked when such paintings and markings have been marred or obliterated.

3.7 **Workmanship.**—All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Lot.**—A lot shall consist of all loading assemblies loaded and assembled by one manufacturer under one contract without change in process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Each lot shall contain—

(a) Metal parts or metal part assemblies from not more than one interfix lot number from one manufacturer.

(b) Loaded illuminant canister assemblies made by one manufacturer at one plant in accordance with the same drawing, same drawing revision, same specification and same specification revision.

(c) Parachute assemblies made by one manufacturer at one plant in accordance with the same drawing, same drawing revision, same specification and same specification revision.

(d) Illuminant composition consisting of batches blended by the same process, each ingredient being from one lot only.

(e) First fire composition consisting of batches blended by the same process, each ingredient being from one lot only.

(f) Black powder from not more than one lot.

4.2 **Sampling.** — Number of samples, acceptance and rejection criteria used for determination of lot acceptance for tests as specified in 4.4.1, 4.5.1, 4.6, and 4.7 shall be in accordance with Standard Inspection Pro-

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cedure ORD–SIP–M18–314. Twenty-two loading assemblies shall be selected by the inspector for test as specified in 4.8. Sampling of component materials is specified in 4.4.2 and 4.5.2 covering test of material.

4.3 Inspection shall be as specified in Specification MIL–G–2550.

4.4 Determination of illuminant composition.

4.4.1 *Moisture content.* — The method of analysis to be used for determining the moisture content of the sodium nitrate in the illuminant composition to determine compliance with 3.4.1, shall be a procedure found applicable to the contracting officer. The contractor shall furnish an affidavit which specifies the determination of moisture content of the sodium nitrate in the illuminant composition.

4.4.2 *Ingredients (Magnesium Only).*

4.4.2.1 *Apparatus.*—Apparatus shall be assembled as shown in Figure 1. The gas evolution apparatus shall consist of a 600 ml Erlenmeyer flask filled with a two-hole rubber stopper. A 25 ml burette containing 6:19 sulfuric acid shall be inserted through one hole of the rubber stopper; the other hole serving to connect the Erlenmeyer gas evolution flask to a V-shaped tube (see 6.2) containing ascarite. This tube is used to absorb any evolved carbon dioxide. A three-way stopcock shall be inserted between the gas burette and the U-tube so that the Erlenmeyer flask can be exposed to atmospheric pressure. The gas burette, calibrated from 450 to 500 ml, shall be connected to a water leveling bulb (see 6.3) filled with water saturated with hydrogen. The gas burette shall be water jacketed to achieve accurate and rapid temperature control. A circulating pump shall be used to circulate the water through the water jacket and a pail of water shall be used as a reservoir. The Erlenmeyer flask shall be immersed in the water in a pail to maintain it at the same temperature as the gas burette.

4.4.2.2 *Procedure.*—Exactly 1.0000 gram of crushed sample shall be transferred to a 600 ml, wide-mouthed Erlenmeyer flask. Two hundred ml of the distilled water saturated with hydrogen shall be added and the Erlenmeyer flask connected to the apparatus. The three-way stopcock shall be opened so that the flask and the gas burette are exposed to atmospheric pressure. The level of the liquid in the gas burette shall be adjusted to the zero point with the leveling bulb. The three-way stopcock shall be turned so that it will only allow gas to flow from the Erlenmeyer flask to the gas burette. The leveling bulb shall be lowered and 25 ml of 6:19 sulfuric acid shall be added from the 25 ml burette. After the addition of the acid, the flask shall be agitated occasionally until the reaction is completed. The reaction shall be considered complete when the volume of generated gas remains within 1 ml for a period of 20 minutes. When the reaction is complete, the level on the gas burette shall be adjusted to that of the water leveling bulb and the volume of gas in the gas burette read. If the evolved gas is less than is measurable by the burette, a known volume of water shall be added through the acid burette unitl a readable volume is reached on the gas burette. The determination must be started and finished at the same temperature (approximately 25°C.). The percentage of magnesium in the sample shall be calculated as follows:

$$\text{Percentage of Magnesium (see 6.4)} = \frac{(A-B)\ (C-D)\qquad .03901}{W(273+T)}$$

Where:

A = volume read on gas burette in ml.

B = volume of acid or volume of acid plus water added from the burette in ml.

C = atmospheric pressure in mm, (Any suitable barometer may be used to determine the pressure.)

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D = vapor pressure of water in mm at the temperature of the determination.

T = temperature at which the determination is made.

W = weight of sample.

4.5 Determination of first fire composition.

4.5.1 *Moisture content.*—The methods of analysis to be used for determining the moisture content of the barium nitrate and zirconium hydride of the first fire composition to determine compliance with 3.4.2, shall be procedures found applicable and acceptable to the contracting officer. The contractor shall furnish an affidavit which specifies the determination of moisture content of the barium nitrate and zirconium hydride in the first fire composition.

4.5.2 *Ingredients.*

4.5.2.1 *Barium Nitrate.* — An accurately weighed portion of exactly 1.0000 gram of the ground sample shall be transferred to a tared medium porosity sintered glass crucible. The contents of the crucible shall be extracted with ten 20-ml portions of distilled water, each portion being stirred and allowed to remain in contact with the residue for approximately 1 minute before suction is applied. The crucible and contents shall be transferred to an oven at 100° to 105°C. for 1 hour, cooled to room temperature in a desiccator, and weighed. The crucible and contents shall be reserved for the tetranitrocarbazole determination (see 4.5.2.2). The percentage of barium nitrate in the mixture shall be calculated as follows:

$$\text{Percentage of Barium Nitrate} = \frac{100A}{W}$$

Where:

A = loss in weight of the crucible and contents.

W = weight of sample.

4.5.2.2 *Tetranitrocarbazole.*—The contents of the crucible reserved from the barium nitrate procedure (see 4.5.2.1) shall be extracted with three 20-ml portions of hot acetone. Each portion shall be stirred and allowed to remain in contact with the residue for approximately 1 minute before applying suction. The crucible and contents shall be transferred to an oven at 100° to 105°C. for 1 hour, cooled to room temperature in a desiccator, and weighed. The crucible and contents shall be reserved for the silicon determination (see 4.5.2.3). The percentage of tetranitrocarbazole in the mixture shall be calculated as follows:

$$\text{Percentage of Tetranitrocarbazole} = \frac{100A}{W}$$

Where:

A = loss in weight of the crucible and contents.

B = weight of sample.

4.5.2.3 *Silicon.*—The crucible and contents reserved from the tetranitrocarbazole procedure (see 4.5.2.2) shall be transferred to a 250 ml beaker, 100 ml of 50 percent potassium hydroxide solution (50 grams KOH/100 ml water) added, the beaker covered with a watchglass and transferred to a steam bath. The crucible shall be heated for 1 hour with occasional swirling, removed from the steam bath and allowed to cool to room temperature. The crucible shall be removed carefully from the beaker with the aid of forceps and washed thoroughly with approximately 50 ml of distilled water. The contents of the beaker shall be filtered quantitatively using the removed crucible. The residue in the crucible shall be washed with ten 20-ml portions of distilled water. The crucible and contents shall be transferred to an oven at 100° to 105°C. for 1 hour, cooled to room temperature in a desiccator, and weighed. The crucible and contents shall be retained for the zirconium hydride determination (see 4.5.2.4). The percentage of silicon in the mixture shall be calculated as follows:

$$\text{Percentage of Silicon} = \frac{.9240A\ (100)}{W}$$

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Where:

A = loss in weight of the crucible and contents.
W = weight of sample.

The factor of .9240 represents a correction factor for the solubility in 50 percent potassium hydroxide solution of laminac Type B and impurities in the zirconium hydride. This figure shall be used except in the event of noncompliance, dispute, or question. In this event, the weight of laminac Type A and impurities dissolved in the 50 percent potassium hydroxide solution shall be determined and the correction factor to be used calculated as follows:

Exactly 0.1500 gram of zirconium hydride, 0.2000 gram of silicon and 0.0500 grams of laminac Type B of the specified grades shall be transferred to a medium porosity sintered glass crucible. The determination shall proceed exactly as in the above silicon determination and the correction factor calculated as follows:

$$\text{Correction factor} = \frac{0.2000}{A}$$

Where:

A = loss in weight of sample.

4.5.2.4 *Zirconium hydride.* — The crucible and contents reserved from the silicon procedure (see 4.5.2.3) shall be transferred to a 250 ml beaker. Fifty ml of concentrated hydrochloric acid and 25 ml concentrated nitric acid shall be added. The beaker shall be covered with a watch-glass and 50 ml of concentrated sulfuric acid shall be added dropwise with caution. The beaker and contents shall be transferred to a hot plate, boiled gently until white sulfur trioxide fumes appear, an additional 75 ml of concentrated sulfuric acid added, and the heating procedure continued until a volume of approximately 25 ml remains. During this heating period, several drops of concentrated nitric acid shall be added cautiously under a watch-glass from time to time until the laminac is entirely decomposed, as evidenced by the permanent clarity of the solution. The beaker and contents shall be cooled to room temperature and sufficient distilled water to make a volume of approximately 100 ml, shall be added cautiously under a watch-glass. The crucible shall be removed from the solution with the aid of forceps and washed quantitatively into the beaker with approximately 50 ml of distilled water. The solution shall be stirred thoroughly with a glass rod and allowed to cool to room temperature. The solution shall be filtered, using a Whatman No. 42 paper, or approved substantial equal, and the filtrate collected in a clean 500 ml beaker. Any residue in the filter paper shall be washed with sufficient distilled water to make the total volume in the 400 ml beaker approximately 200 ml. The filter paper shall be discarded and the beaker cooled in an ice bath to 10° ± 2°C. To the beaker in the ice bath shall be added, with stirring, 20 ml of freshly prepared cupferron solution (dissolve 6 grams of cupferron in 100 ml of cold water and filter). The precipitate shall be filtered using a Whatman No. 40 paper, or substantial equal, and washed with five 20 ml portions of cold 1N (1:10) hydrochloric acid. The precipitate and paper shall transferred to a tared crucible and dried in an oven at 100° to 105°C. for 2 hours. The crucible and contents shall be placed in a muffle furnace at 300° to 400°C. until charring of the paper has occurred and the rush of gases from decomposition of the organic matter has ceased. The temperature shall be increased to 900° ± 50°C. and the ignition continued for ½ hour. The crucible and contents shall be cooled to room temperature in a desiccator and weighed as zirconium oxide. The percent of zirconium hydride in the mixture shall be calculated as follows:

$$\text{Percent of Zirconium Hydride} = \frac{75.65A}{W}$$

Where:

A = gain in weight of the crucible.
W = weight of sample.

4.5.2.5 *Laminac, Type B.*—The percentage of laminac, Type B in the mixture shall be

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calculated by subtracting from 100 the combined percentages of barium nitrate, silicon, tetranitrocarbazole and zirconium hydride.

4.6 Determination of moisture content of black powder.—The allowable moisture content and the method of determining the percentage thereof shall be in accordance with Specification JAN–P–223 to determine compliance with 3.4.3.

4.7 Static test of illuminant canister assembly.—Sample illuminant canister assemblies shall be selected and tested to determine the candlepower and burning time. Determination of candlepower shall be in accordance with Specification MIL–P–20464.

4.8 Proving ground test.

4.8.1 *Procedure.* — Twenty-two loading assemblies shall be selected for test, assembled with M501 MTSQ Fuzes and assembled into complete rounds. Fifteen rounds shall be fired from a 105-mm, M2 or M4, Howitzer with the howitzer elevations adjusted to give a height of burst of approximately 2,000 feet. The rounds shall be divided into two groups, one of 5 and the other of 10, and propelling charges and fuze settings shall be in accordance with Tables I and II. The seven remaining rounds shall be held in reserve.

Note. Should a round function below 1,000 feet because of the fuze action, the results should be disregarded and a replacement round fired. However, this shall be limited to not more than one replacement round per each phase.

TABLE I. *Safety phase.*

Number rounds	Propelling charge	Fuze setting
5	112 percent of Charge 7 Pressure	12–20 sec.

TABLE II. *Functioning phase.*

Number rounds	Propelling charge	Fuze setting
5	Charge 7	5–10 sec.
5	Charge 7	12–20 sec.

The necessary observations shall be made and the results obtained from firing each round shall be scored in accordance with Table III to determine compliance with 3.5.1 and 3.5.2.

TABLE III. *Time of effective illumination.*
(See note)

	43 seconds plus	33 to 43 seconds	23 to 33 seconds	13 to 23 seconds	0 to 13 seconds
Score as	100 percent	75 percent	50 percent	25 percent	0 percent

Note. "Effective illumination" shall be defined as burning of the canister with normal candlepower in respect to standard shell without exploding or breaking up while suspended by the parachute (the showering of sparks shall not be considered a defect unless continued and excessive). Burning of the canister after striking the ground shall not be considered effective illumination. Any occurrence which detract from "effective illumination" as herein defined shall be recorded on the firing record (see data recording table, "Malfunctions and Remarks" column).

In conducting the safety and functioning phase of the proving ground test, the following data shall be recorded:

Round No.	Sample No.	Elev.	Fuze Setting (sec.)	Prop. Charge	Time (seconds)				Score	Malfunctions and Remarks
					Approx. yds. to Burst	Eject to Open	Eject to Ground	Effective Illum.		

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The sum total of the scores divided by 15 (less the number of fuze failures, which shall be disregarded) shall be the percentage performance.

4.8.2 *Acceptance.* — The lot shall be considered acceptable provided the rounds fire, no premature bursts occur, and provided a percentage performance of not less than 75 percent is obtained. The lot shall be considered eligible for retest under the provisions of 4.8.3 provided no premature bursts occur and a percentage performance of less than 75 percent but not less than 60 percent is obtained. If the limits of eligibility for retest are exceeded, the lot shall be rejected.

4.8.3 *Retest.* — A retest, as specified in 4.8.2, may be made if requested by the contractor using double the number of samples used in the original test, with the exception that only one replacement round shall be allowed per phase for fuze action causing functioning of the round below 1,000 feet. The lot shall be considered acceptable on retest provided no premature burst occurs and provided the sum total of the scores of the original tests plus retest divided by 45 (less the number of rounds disregarded for fuze failures) is not less than 75 percent.

5. PREPARATION FOR DELIVERY

5.1 **Packing, labeling and marking.**—Packing, labeling and marking shall be as specified on the applicable drawings.

5.2 **Data cards.** — Data card information shall be as specified in Specification MIL–G–2550.

6. NOTES

6.1 **Ordering Data.** — Procurement documents should specify, title, number and date of this specification.

6.2 Shwartz U-shaped glass stoppered Tube, Fisher Catalog No. 9–230, was found to be satisfactory for use.

6.3 Leveling bulb (1-liter capacity) Fisher Catalog No. 10–788 was found to satisfactory for use.

6.4 It is to be noted that the equations for calculation the percentages of the various ingredients present are based upon the assumption that the ingredients are 100 percent on any other purity, a factor can be introduced into the calculation.

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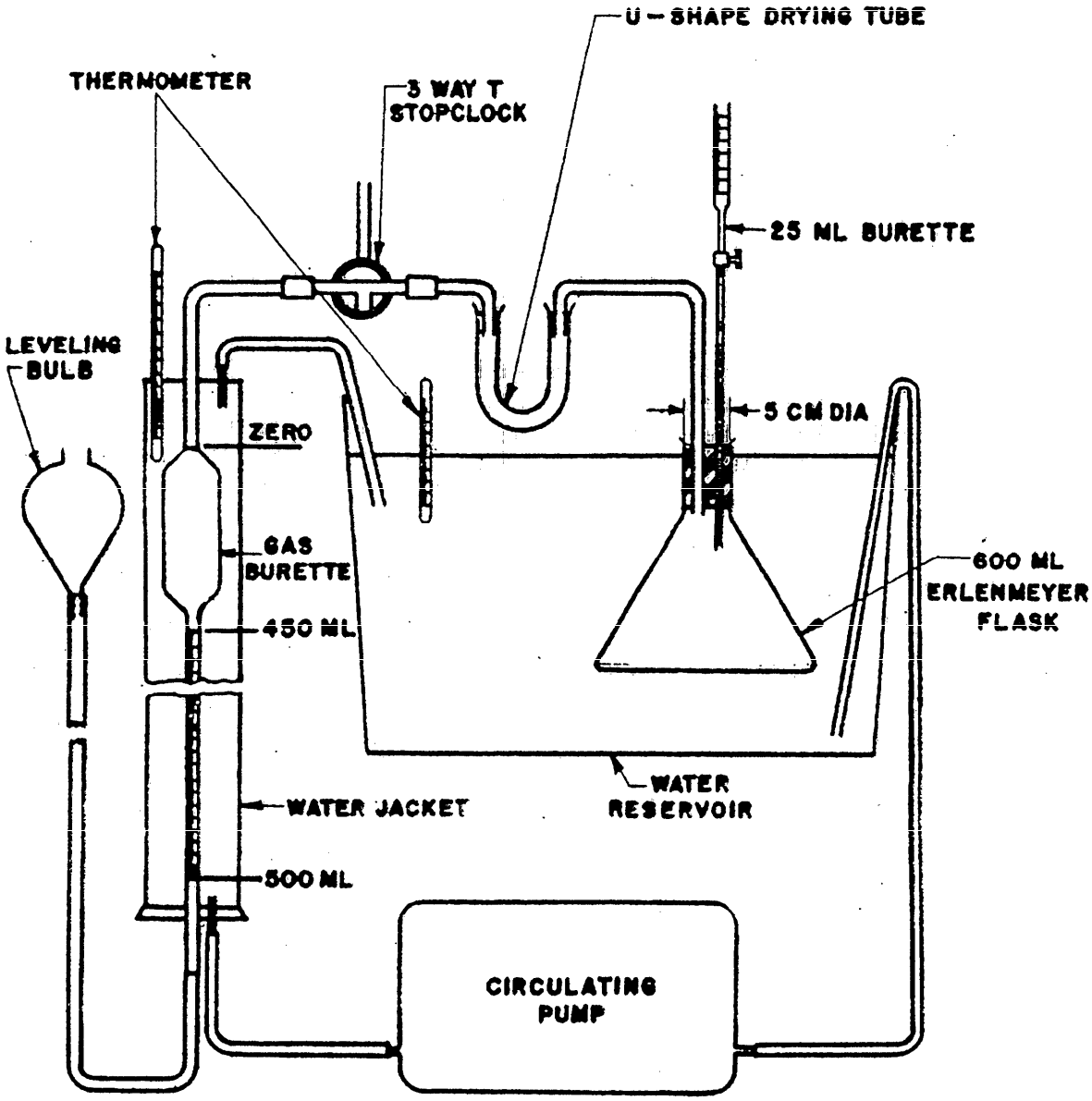


FIGURE 1. *Equipment assembly.*

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